Supplementary Case Atlas for FlexiCT Disagreements

Classification of Abdominal CT Diagnoses Relevant to Emergency Care with Adapted Foundation Models

This atlas contains all 20 disagreements between the full-development FlexiCT final-block ensemble and the original test labels. One radiologist reviewed the cases with access to model predictions; clinical records were consulted where available. Original labels were retained for all primary performance measures. Hypothetical scenario assignments are detailed in Tables S6 and S7 of the Supplementary Materials and Methods; no scenario replaces the original reference labels in the primary analysis.

Each case includes axial, coronal, and sagittal CT views, predicted- and reference-class attribution maps, their signed contrast, and an axial CT overview. CT is displayed at window level 40 HU and width 400 HU. Positive maps are normalized separately from 0 to 1. Red in the signed contrast favors the predicted class and blue the reference class. Planes pass through the predicted-class attribution maximum, without lesion annotations. The overview samples the resampled volume and is not a set of lesion-selected slices.

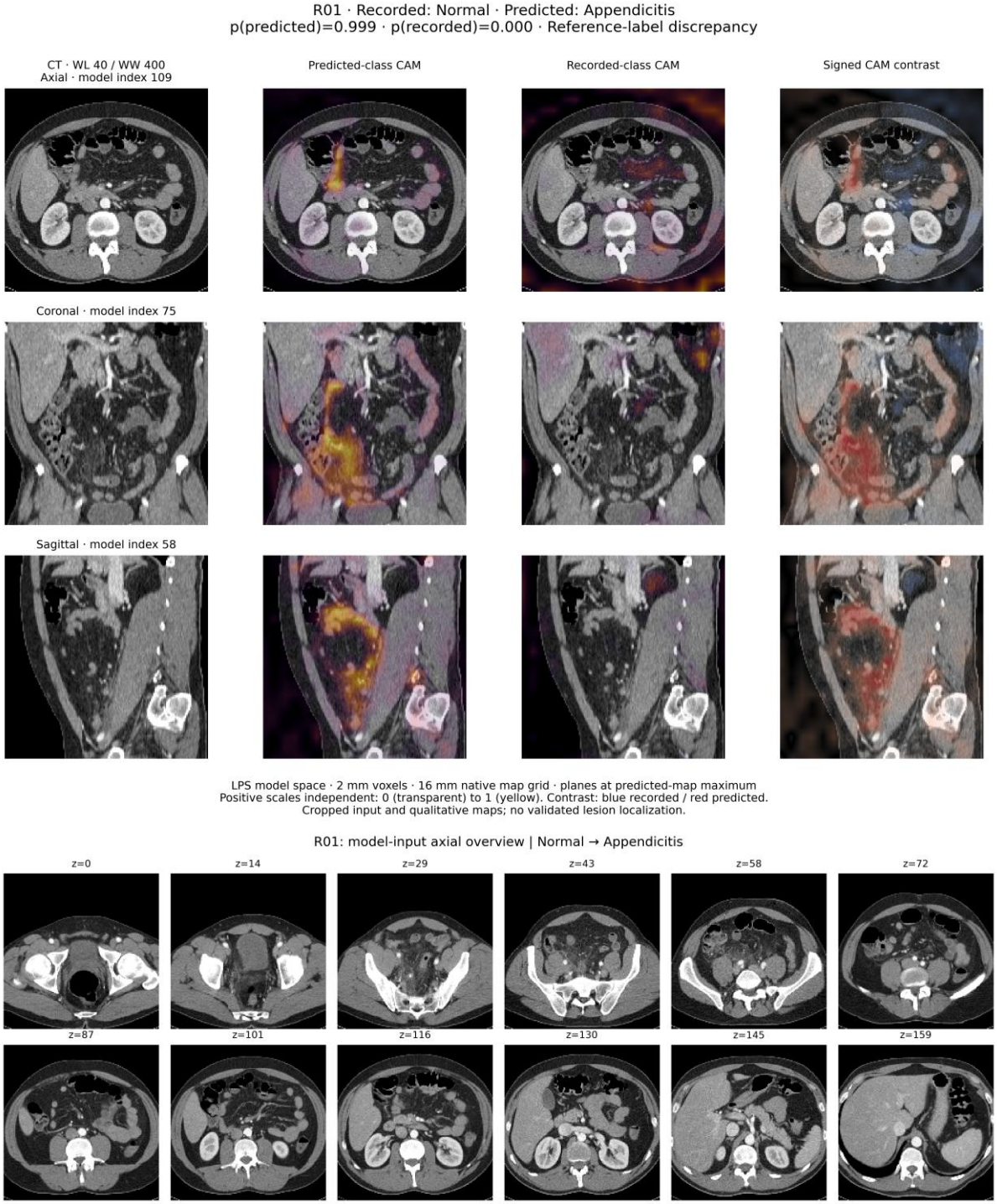
Attribution was calculated on a 20^3 grid from a 160^3 input at 2-mm spacing. Occlusion masked six 40^3 cubes (9.375% of the input). Targeted and three random masks used zero standardized intensity; the minimum-intensity control used the minimum input value. Δp denotes the original predicted-class probability minus the post-occlusion probability; positive values indicate a reduction. Correlations are Spearman coefficients. The randomization control changes only the final classifier layer of the first fold. These exploratory controls do not establish lesion localization.

In the image panels, p denotes predicted-class probability, z/y/x denote resampled slice indices, and CAM denotes class activation map. Model coordinates use left-posterior-superior orientation. Each original reference and model prediction is shown unchanged. R20 has a provisional clinical-comment assignment requiring confirmation.

R01 Reference-label discrepancy

Original reference: Normal. FlexiCT prediction: Appendicitis (probability 0.999; 5/5 fold votes).

Reviewer comment: Appendicitis identified on reread, consistent with the model prediction.

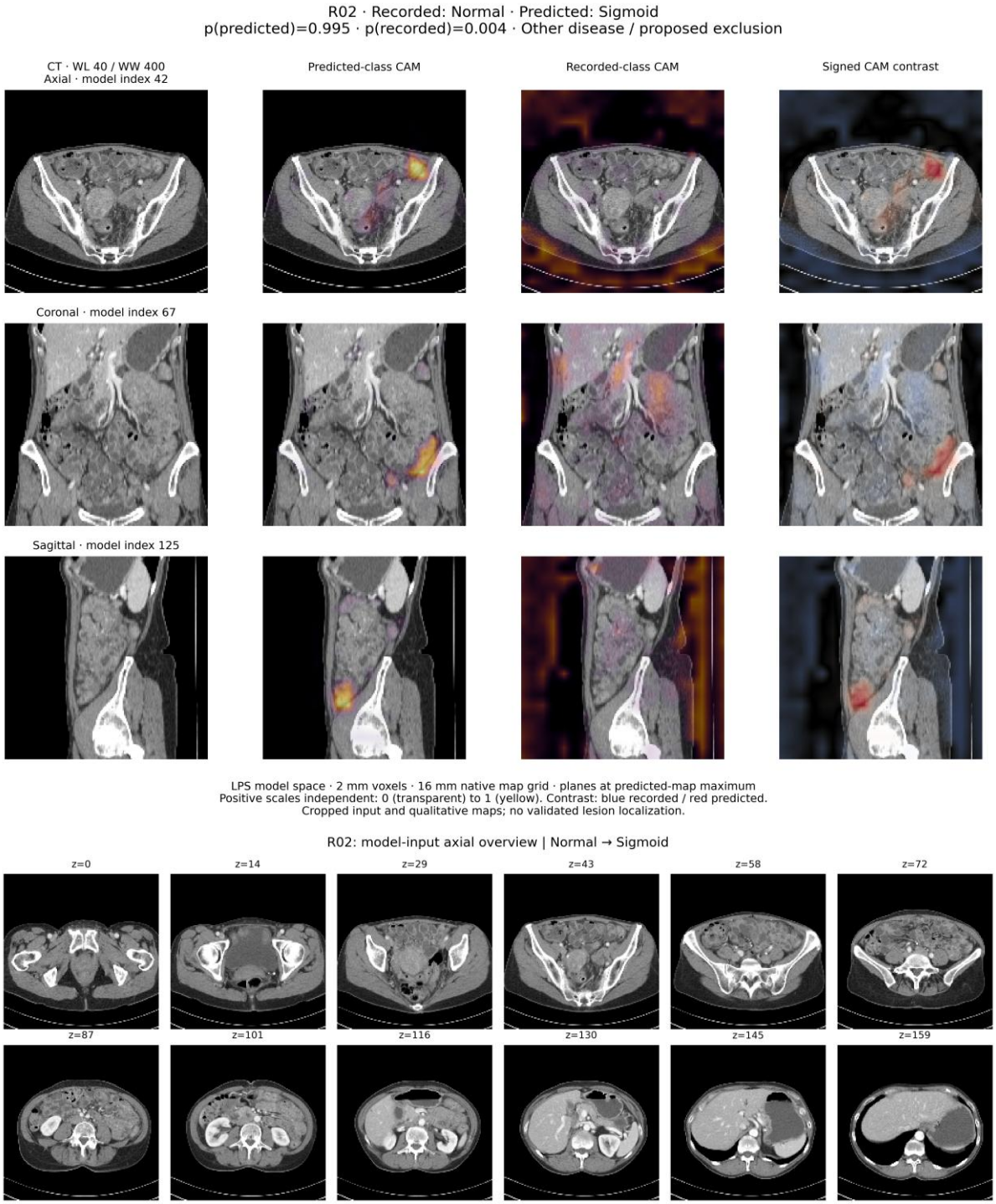


Occlusion Δp: targeted 0.987; random mean 0.030; minimum-intensity control 0.340. Fold-map correlation: 0.594; randomized-head correlation: -0.101.

R02 Other disease and proposed exclusion

Original reference: Normal. FlexiCT prediction: Sigmoid diverticulitis (probability 0.995; 5/5 fold votes).

Reviewer comment: Colitis and jejunitis; neither normal nor appendicitis nor sigmoid diverticulitis.

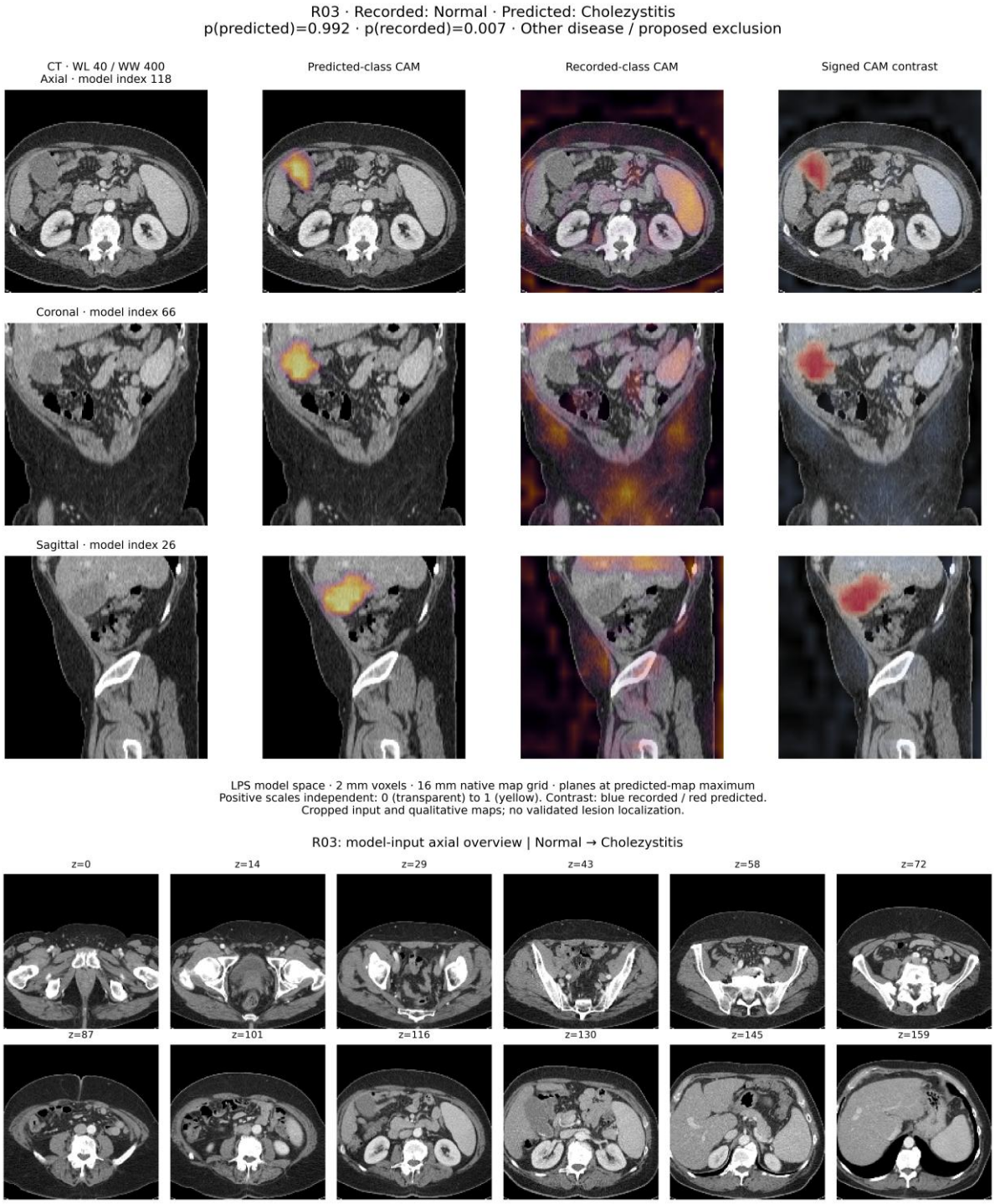


Occlusion Δp: targeted 0.994; random mean 0.341; minimum-intensity control 0.995. Fold-map correlation: 0.536; randomized-head correlation: 0.025.

R03 Other disease and proposed exclusion

Original reference: Normal. FlexiCT prediction: Cholecystitis (probability 0.992; 5/5 fold votes).

Reviewer comment: Enhancing lesion adjacent to the gallbladder, subsequently considered a liver tumor mimicking inflammation.

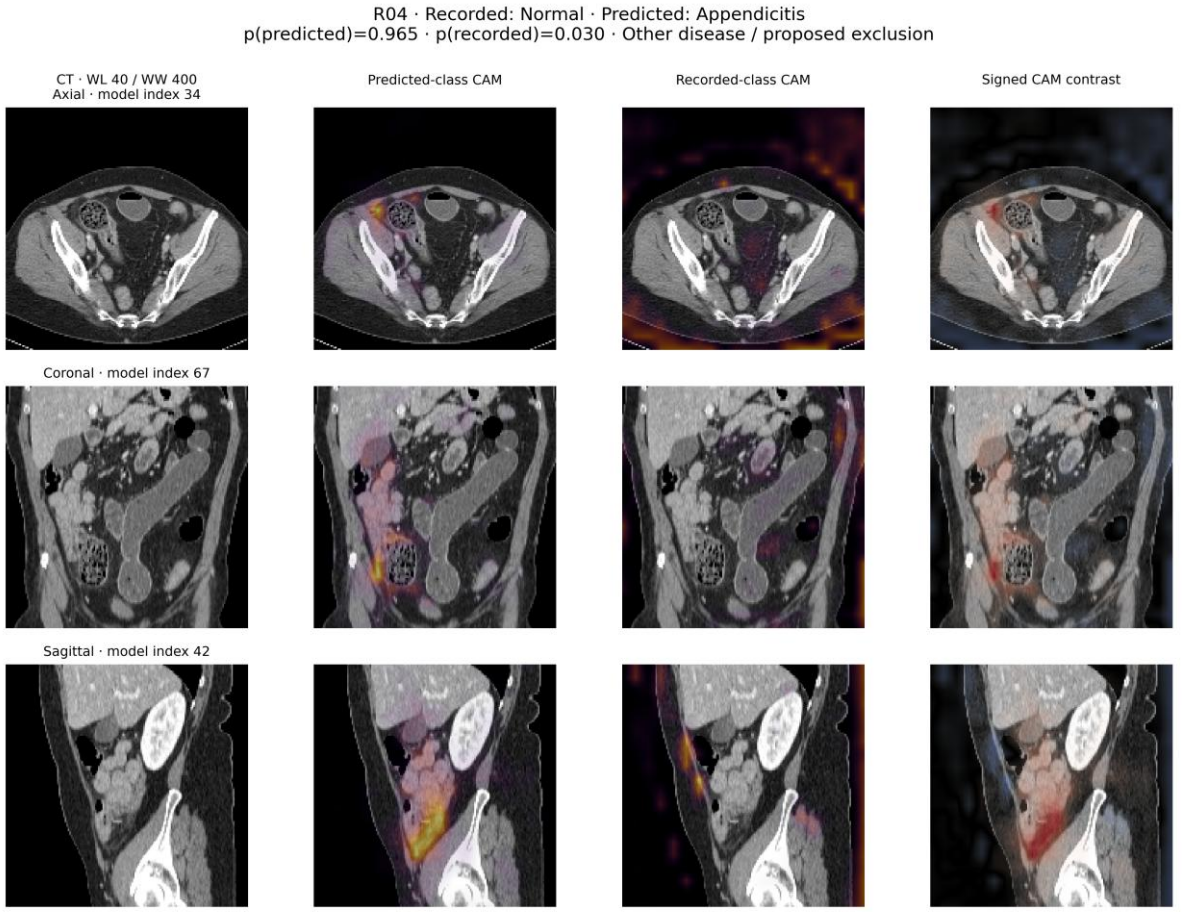


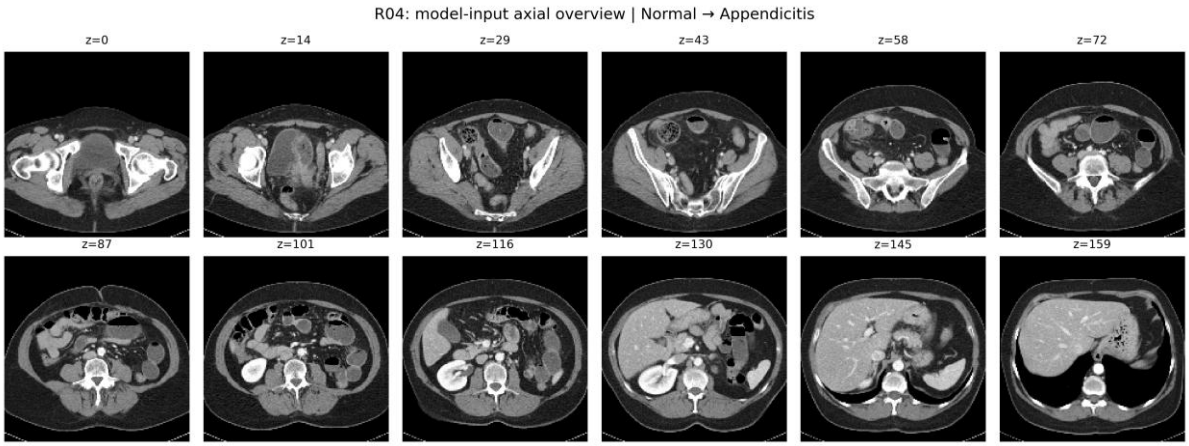
Occlusion Δp: targeted 0.991; random mean 0.025; minimum-intensity control 0.991. Fold-map correlation: 0.597; randomized-head correlation: 0.003.

R04 Other disease and proposed exclusion

Original reference: Normal. FlexiCT prediction: Appendicitis (probability 0.965; 5/5 fold votes).

Reviewer comment: Crohn disease exacerbation with surrounding inflammation and tumor infiltration.



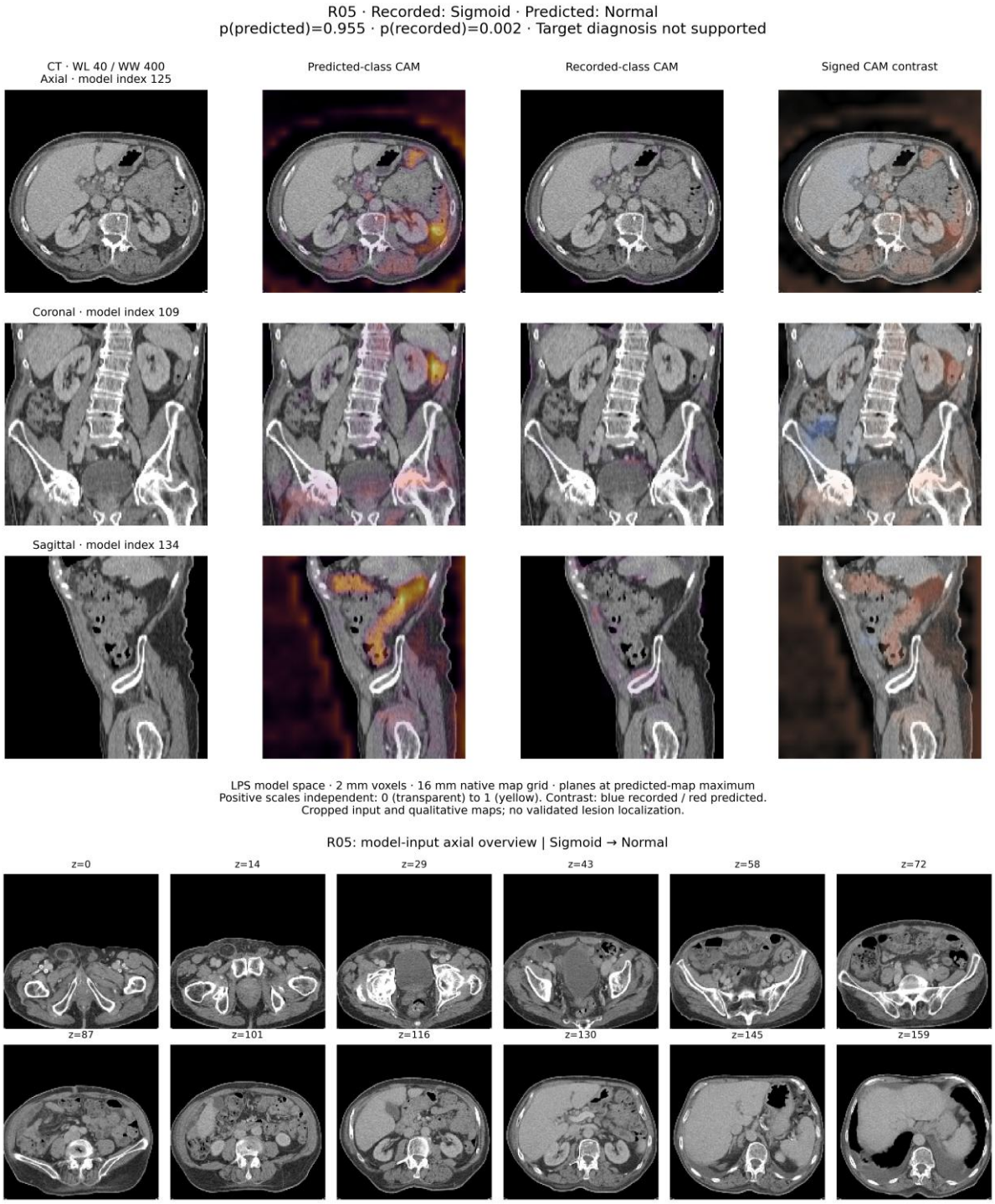


Occlusion Δp: targeted 0.963; random mean 0.535; minimum-intensity control 0.963. Fold-map correlation: 0.662; randomized-head correlation: -0.064.

R05 Target diagnosis not supported

Original reference: Sigmoid diverticulitis. FlexiCT prediction: Normal (probability 0.955; 5/5 fold votes).

Reviewer comment: Incarcerated left inguinal hernia involving sigmoid; no clinical or operative proof of diverticulitis.

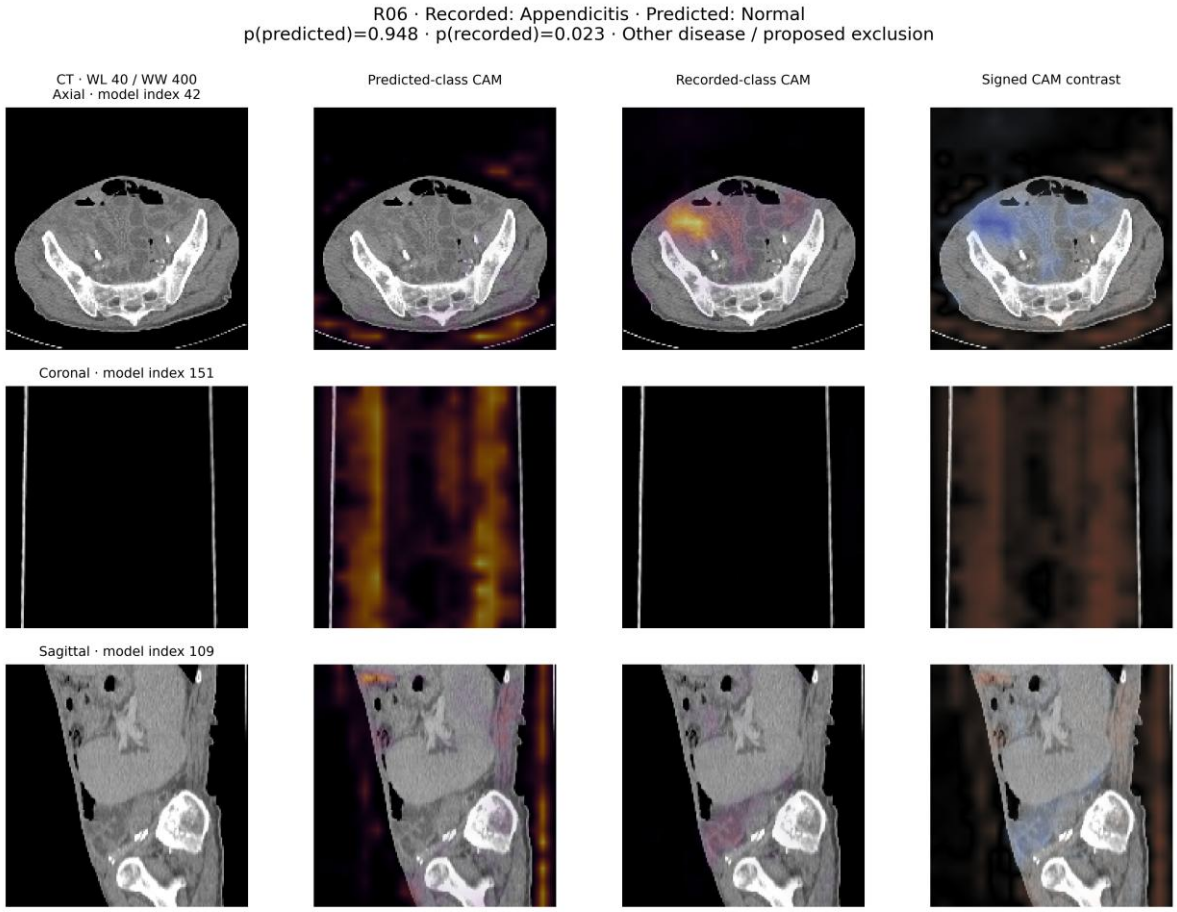


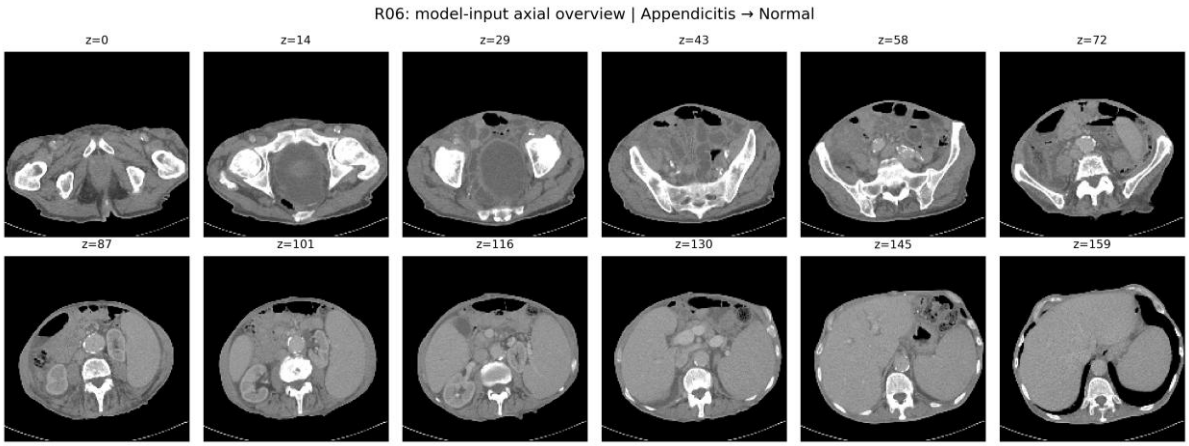
Occlusion Δp: targeted -0.003; random mean -0.041; minimum-intensity control 0.002. Fold-map correlation: 0.841; randomized-head correlation: -0.528.

R06 Other disease and proposed exclusion

Original reference: Appendicitis. FlexiCT prediction: Normal (probability 0.948; 5/5 fold votes).

Reviewer comment: Malignancy-associated rectal obstruction, splenomegaly and ascites; no appendicitis.



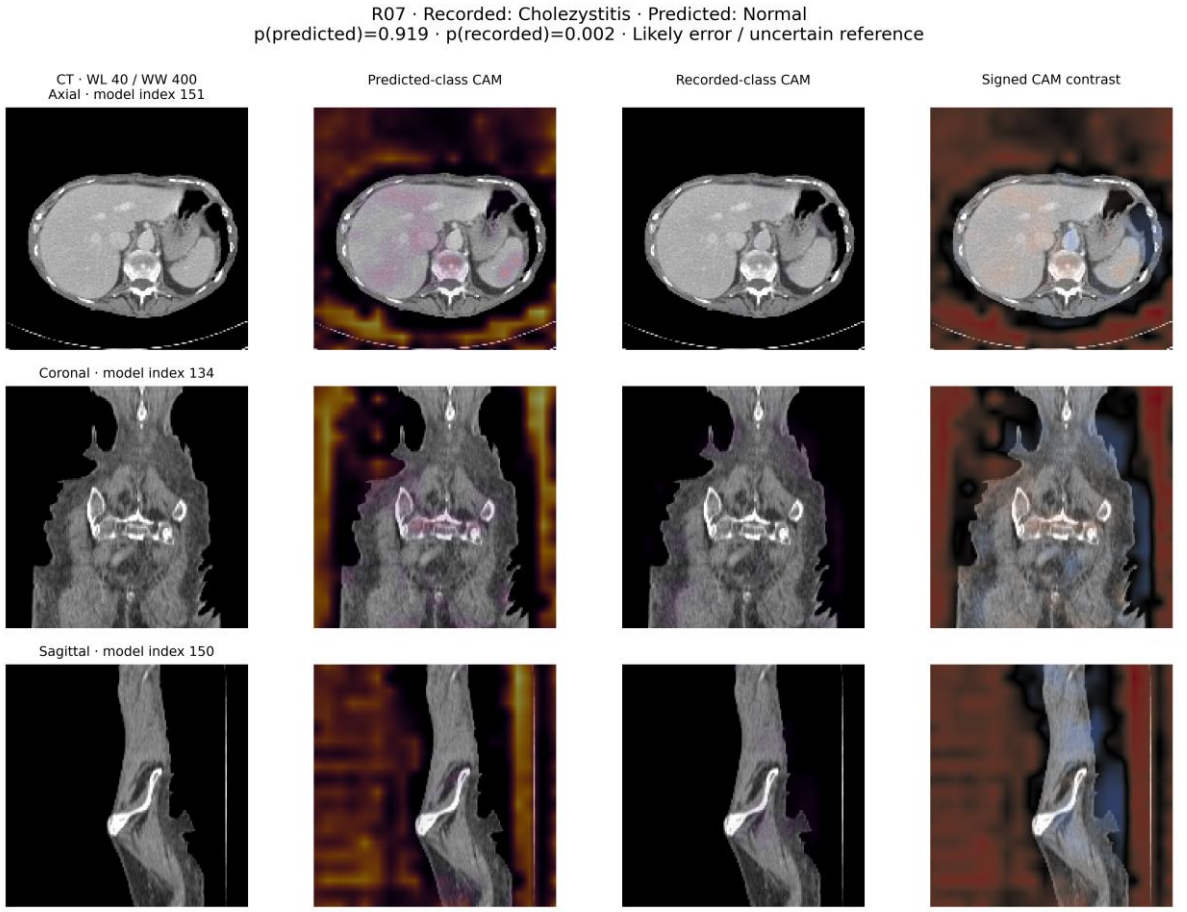


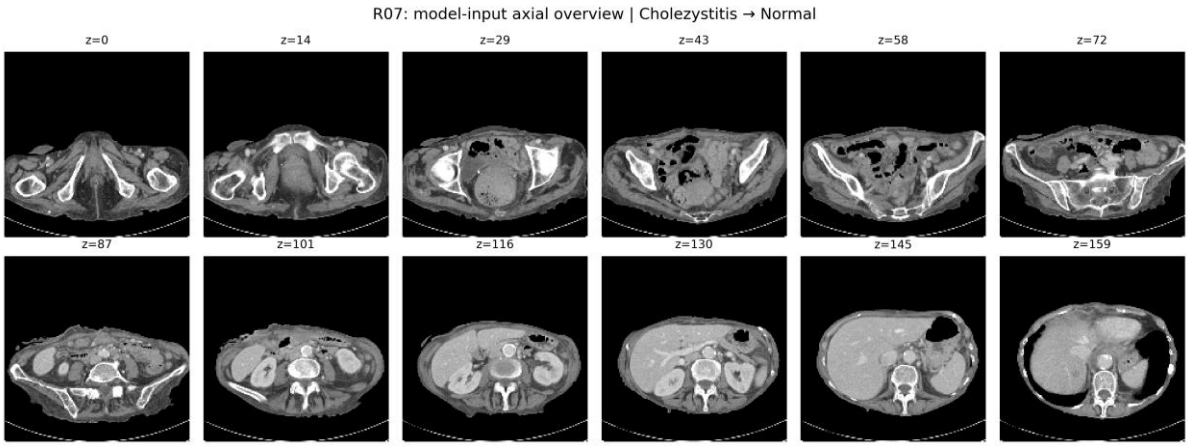
Occlusion Δp: targeted -0.007; random mean -0.037; minimum-intensity control 0.002. Fold-map correlation: 0.739; randomized-head correlation: -0.369.

R07 Likely error and uncertain reference

Original reference: Cholecystitis. FlexiCT prediction: Normal (probability 0.919; 5/5 fold votes).

Reviewer comment: Cholecystitis likely but not proven; nearby postoperative stoma may mimic inflammation.



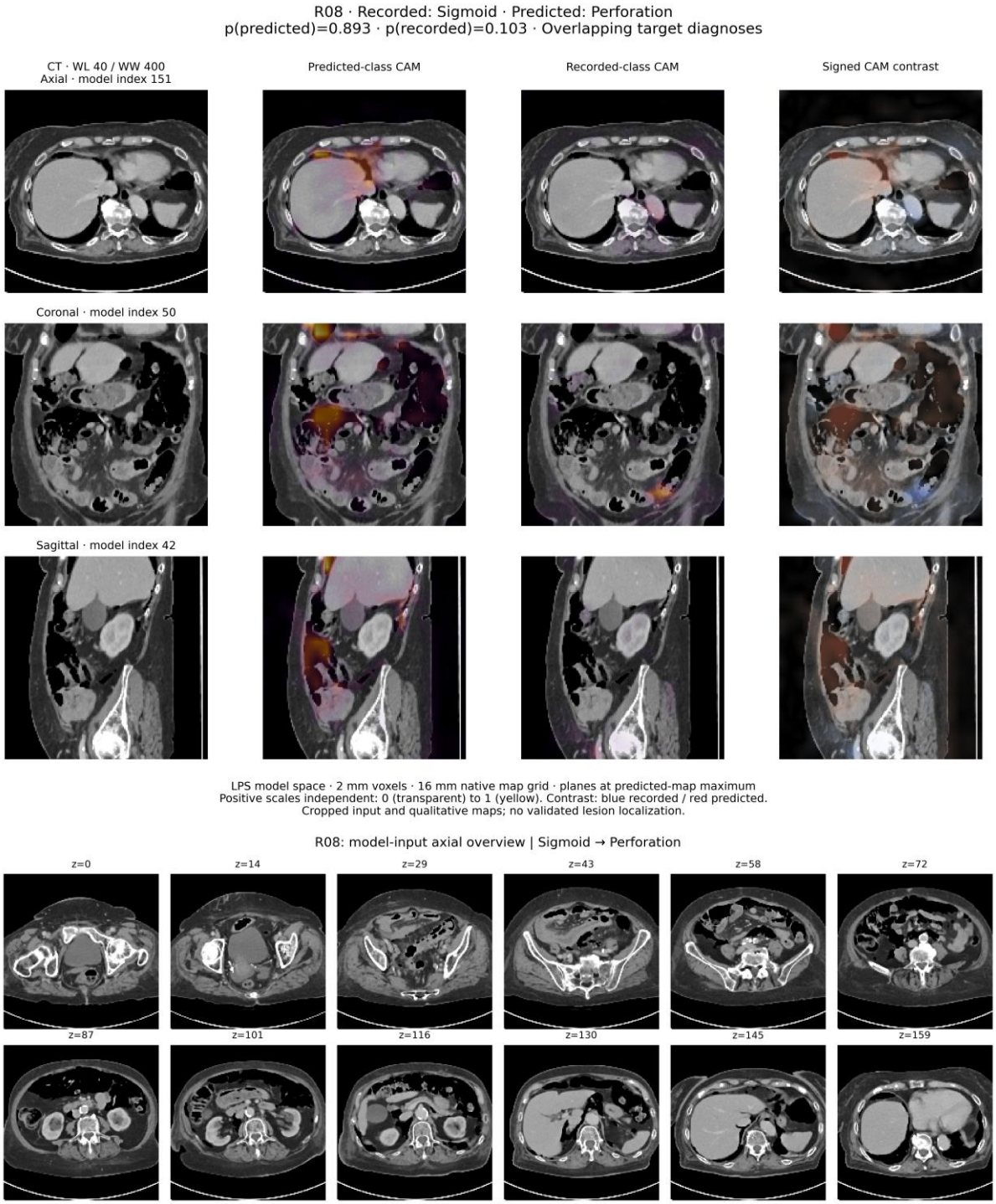


Occlusion Δp: targeted -0.025; random mean -0.037; minimum-intensity control 0.009. Fold-map correlation: 0.828; randomized-head correlation: -0.211.

R08 Overlapping target diagnoses

Original reference: Sigmoid diverticulitis. FlexiCT prediction: Perforation (probability 0.893; 5/5 fold votes).

Reviewer comment: Acute and chronic sigmoid diverticulitis with perforation.

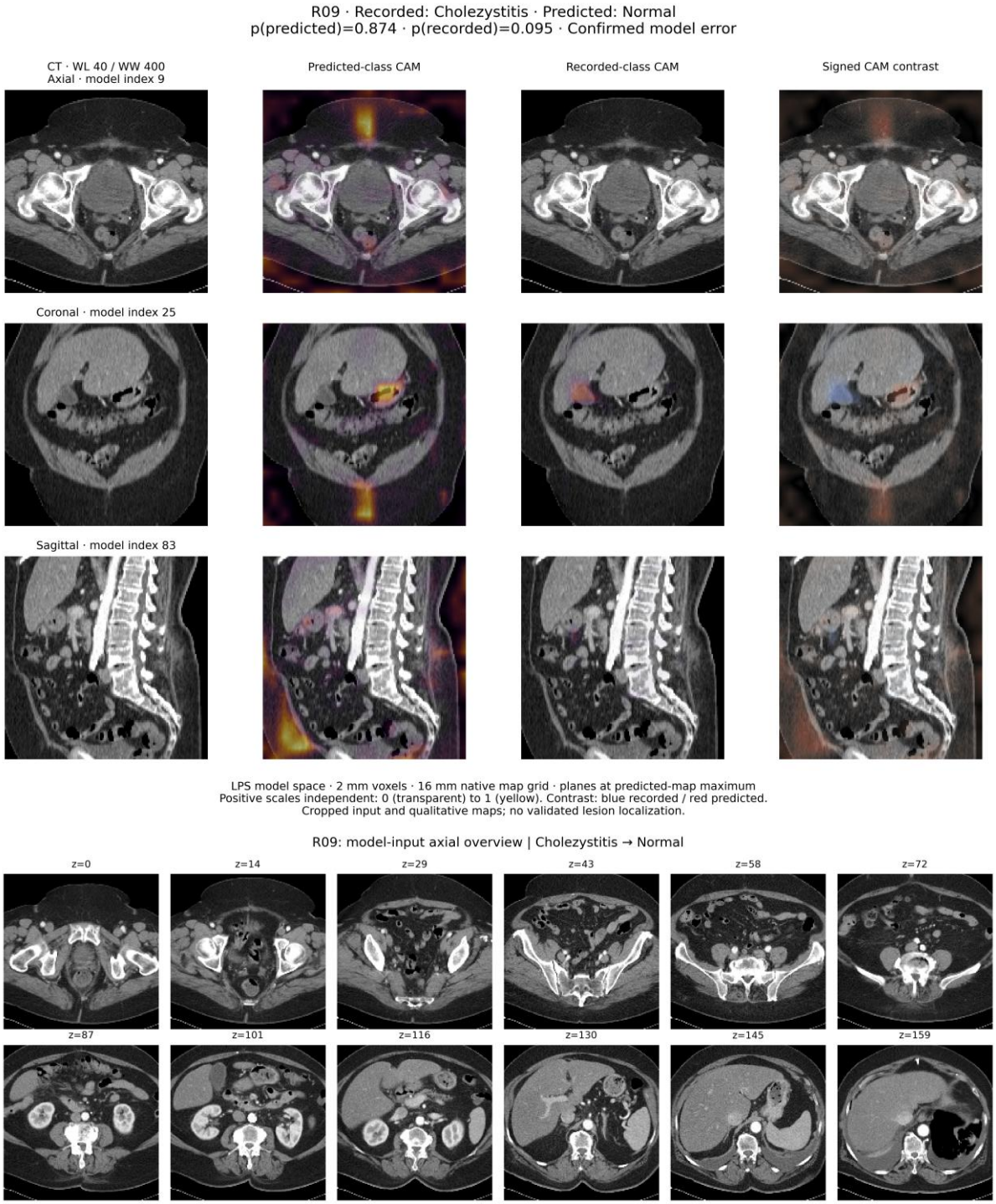


Occlusion Δp: targeted -0.019; random mean -0.042; minimum-intensity control -0.052. Fold-map correlation: 0.573; randomized-head correlation: -0.283.

R09 Confirmed model error

Original reference: Cholecystitis. FlexiCT prediction: Normal (probability 0.874; 5/5 fold votes).

Reviewer comment: Subtle cholecystitis with clinical confirmation and cholestasis.

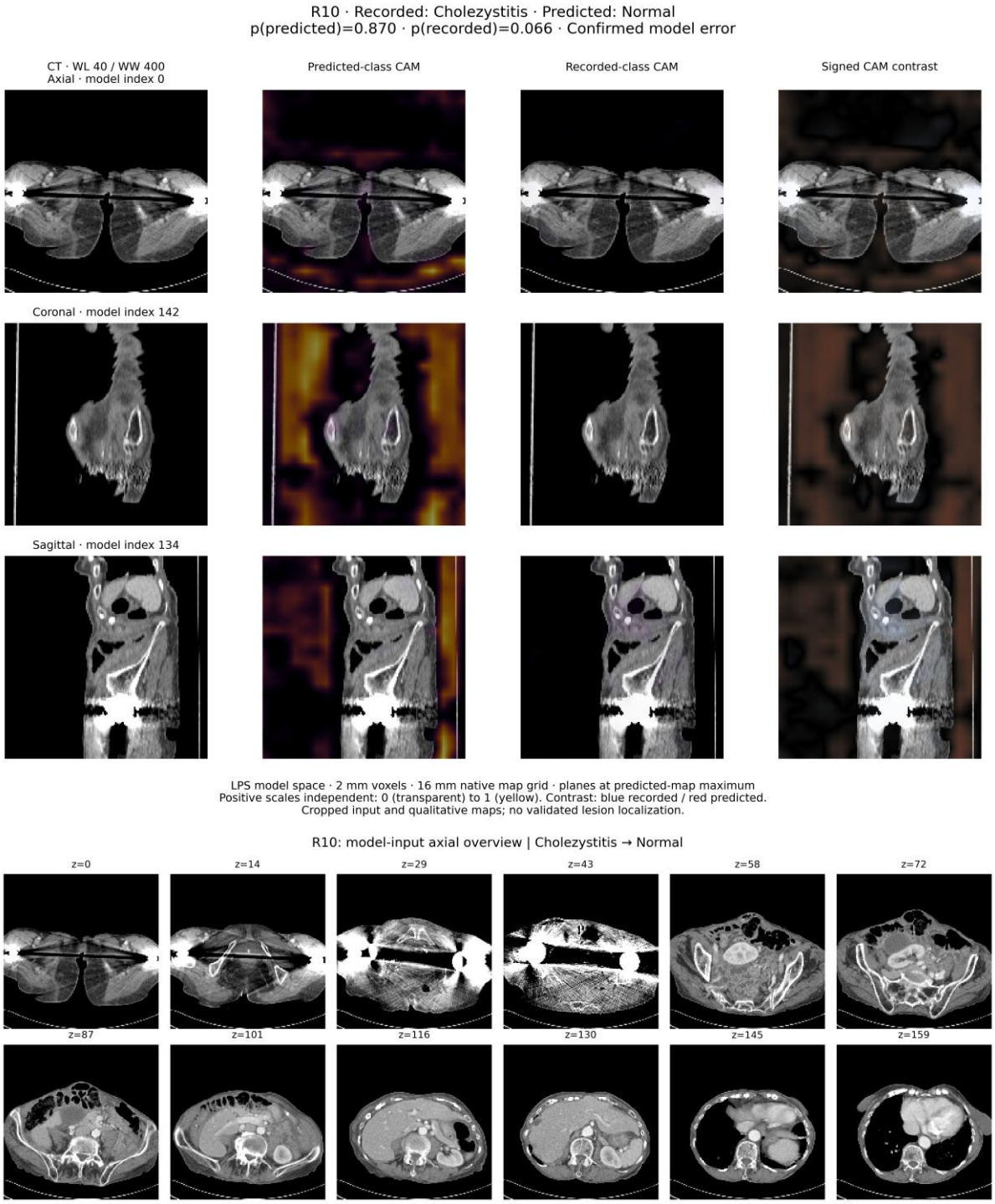


Occlusion Δp: targeted -0.097; random mean -0.110; minimum-intensity control -0.033. Fold-map correlation: 0.719; randomized-head correlation: -0.277.

R10 Confirmed model error

Original reference: Cholecystitis. FlexiCT prediction: Normal (probability 0.870; 5/5 fold votes).

Reviewer comment: Confirmed cholecystitis with a displaced kidney, hernias, sarcopenia, and chronic inflammation.

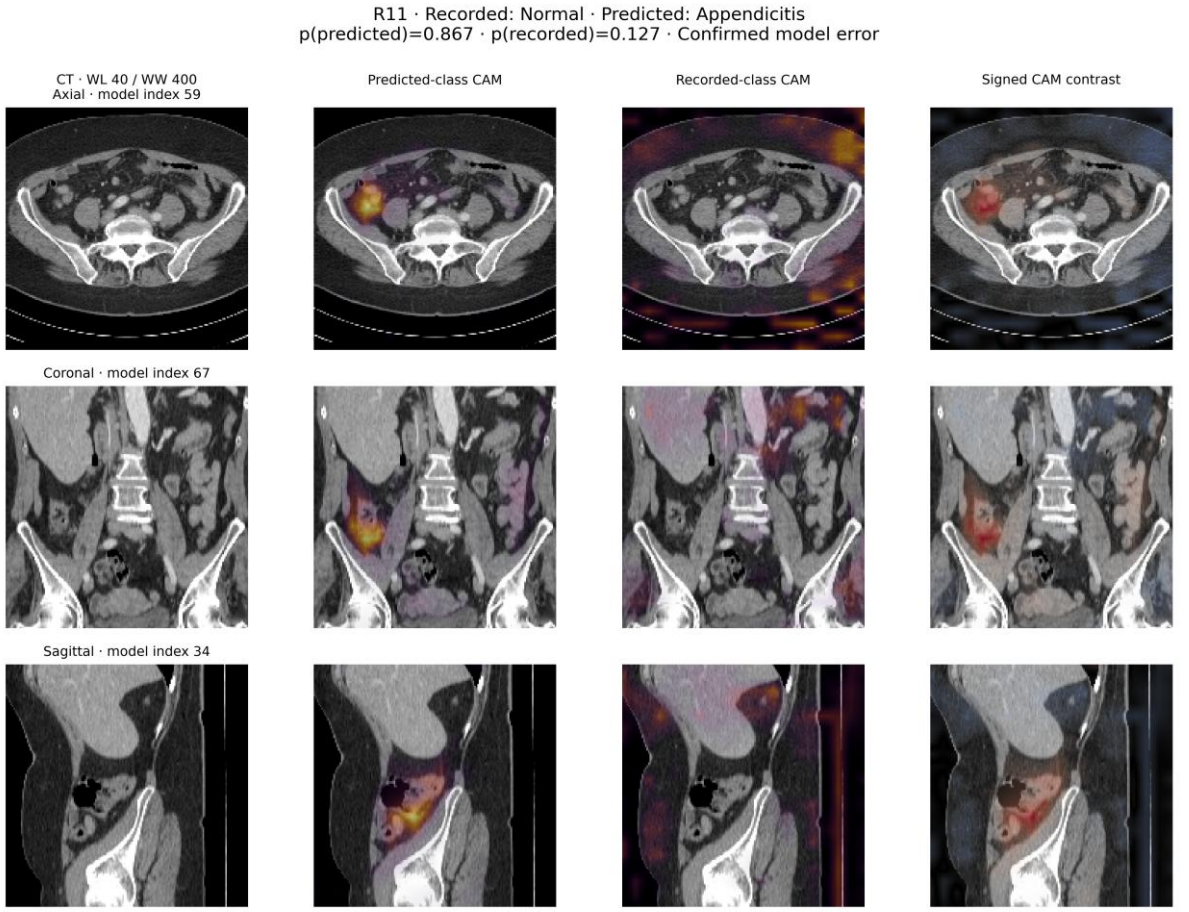


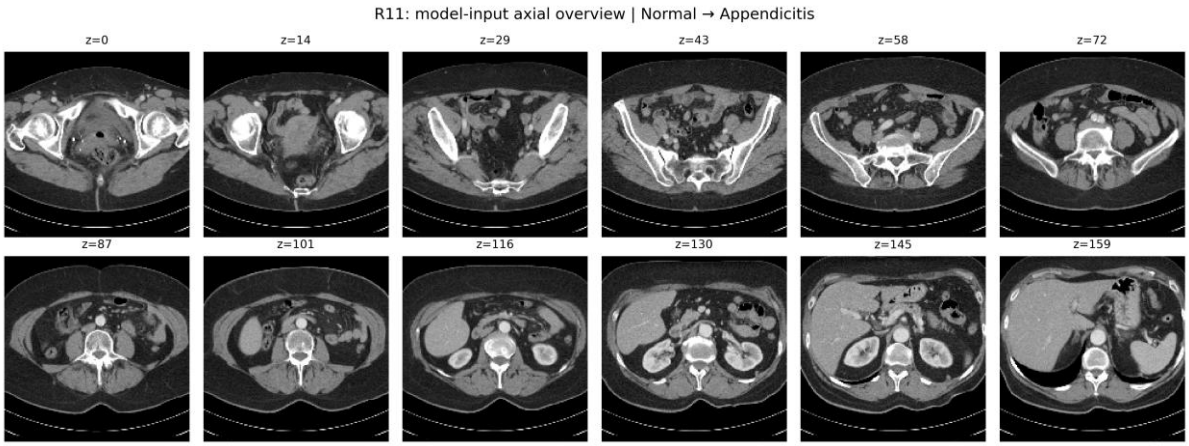
Occlusion Δp: targeted 0.123; random mean 0.022; minimum-intensity control 0.033. Fold-map correlation: 0.827; randomized-head correlation: -0.264.

R11 Confirmed model error

Original reference: Normal. FlexiCT prediction: Appendicitis (probability 0.867; 5/5 fold votes).

Reviewer comment: No abnormality identified on either review.



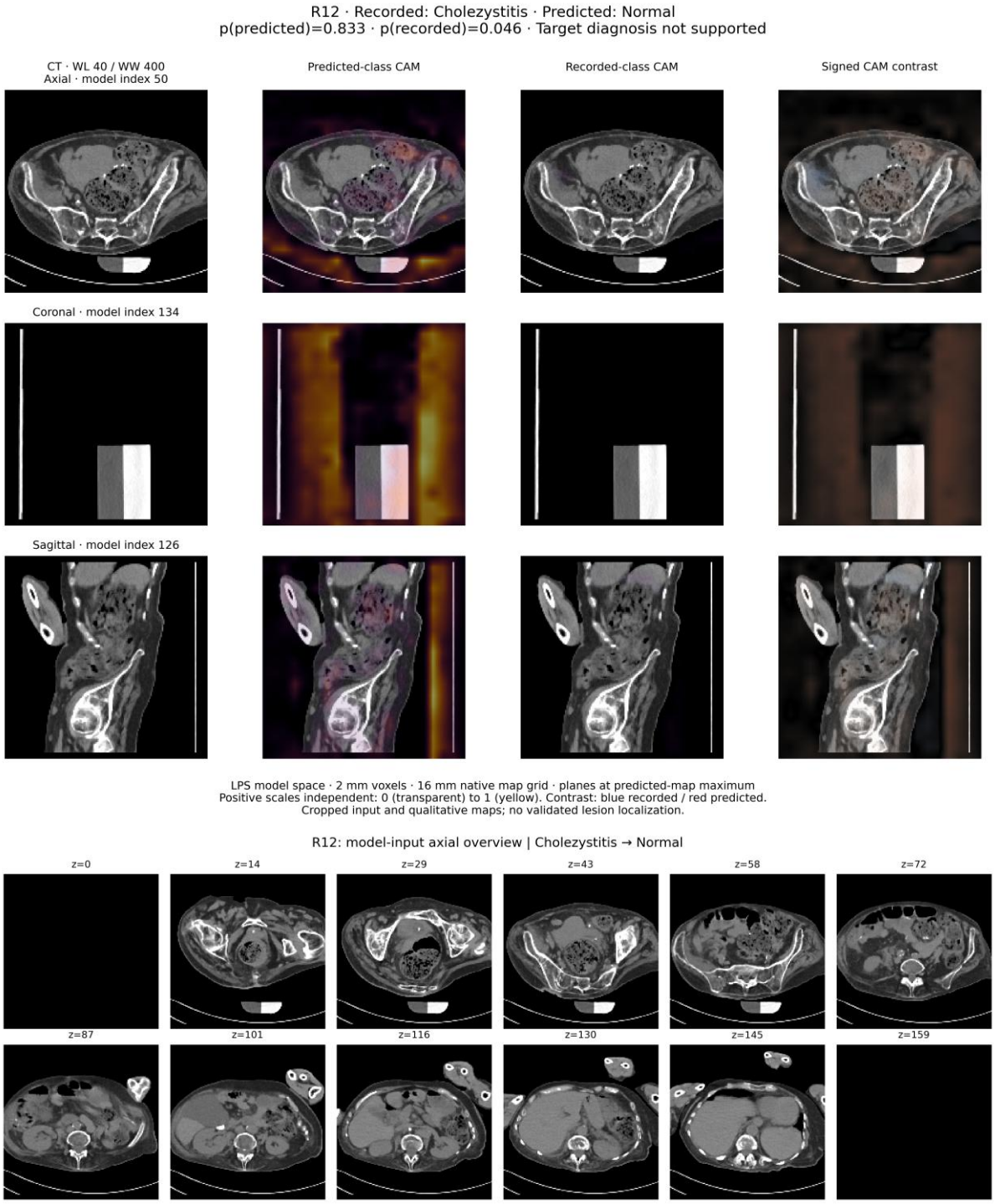


Occlusion Δp: targeted 0.867; random mean 0.866; minimum-intensity control 0.866. Fold-map correlation: 0.717; randomized-head correlation: 0.093.

R12 Target diagnosis not supported

Original reference: Cholecystitis. FlexiCT prediction: Normal (probability 0.833; 5/5 fold votes).

Reviewer comment: Urosepsis and gallbladder hydrops; initial noncontrast CT diagnosis of cholecystitis was not supported.

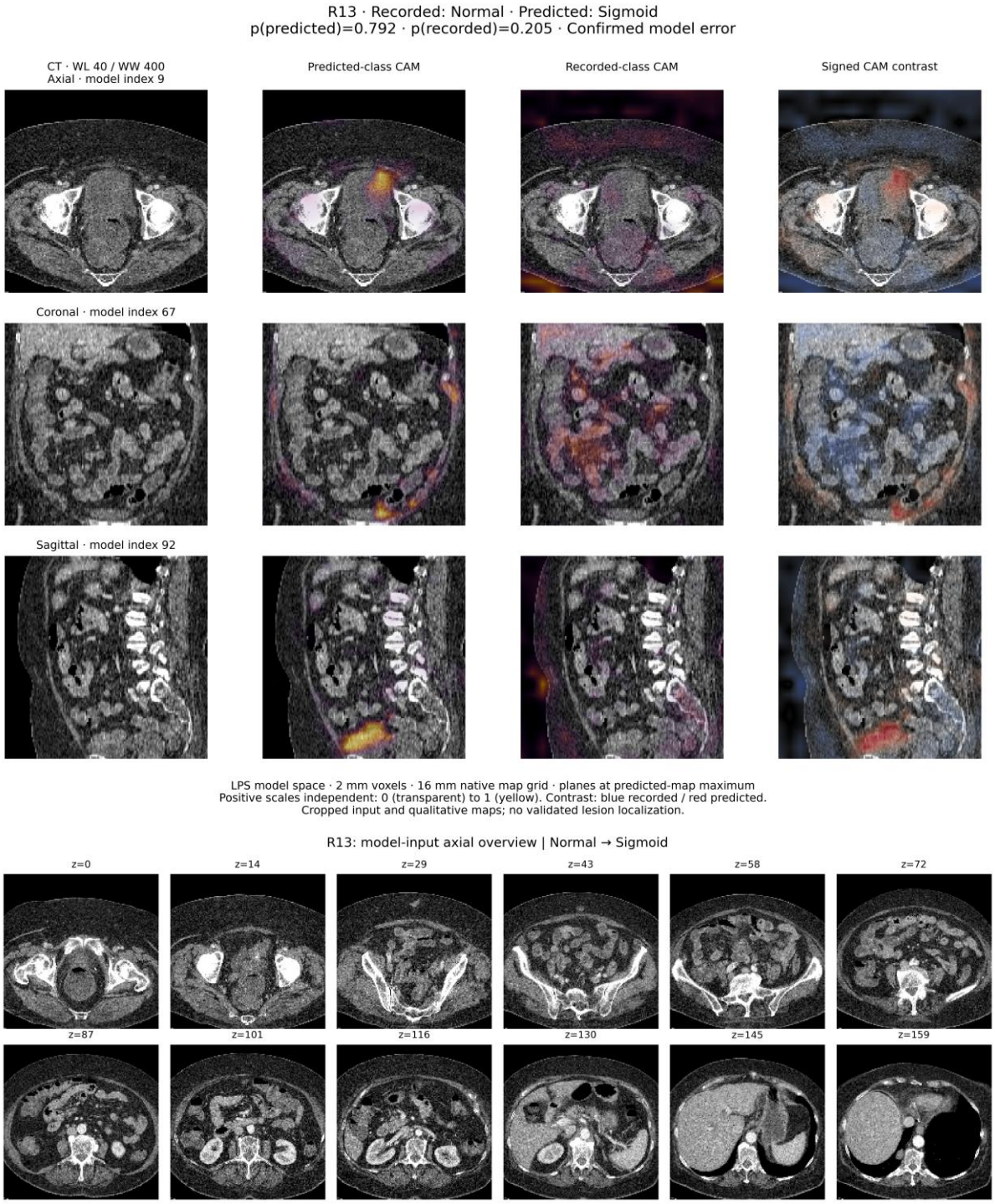


Occlusion Δp: targeted 0.151; random mean 0.051; minimum-intensity control 0.067. Fold-map correlation: 0.629; randomized-head correlation: 0.167.

R13 Confirmed model error

Original reference: Normal. FlexiCT prediction: Sigmoid diverticulitis (probability 0.792; 5/5 fold votes).

Reviewer comment: No abnormality identified on two reviews; a dense sigmoid clip was considered a possible mimic.

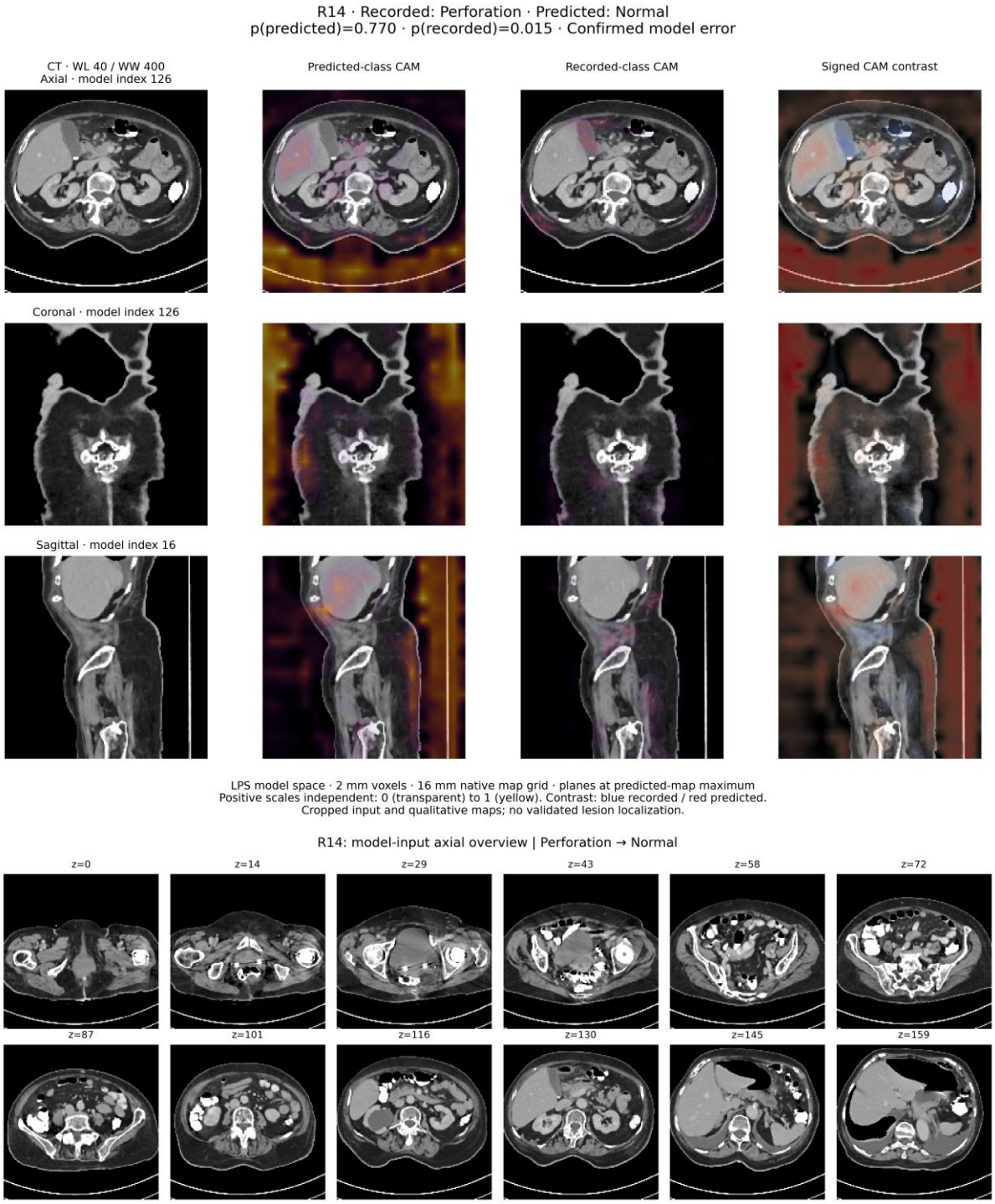


Occlusion Δp: targeted 0.770; random mean 0.652; minimum-intensity control 0.739. Fold-map correlation: 0.564; randomized-head correlation: -0.053.

R14 Confirmed model error

Original reference: Perforation. FlexiCT prediction: Normal (probability 0.770; 5/5 fold votes).

Reviewer comment: Surgically confirmed contained duodenal perforation with a large gas–fluid collection and atypical anatomy.

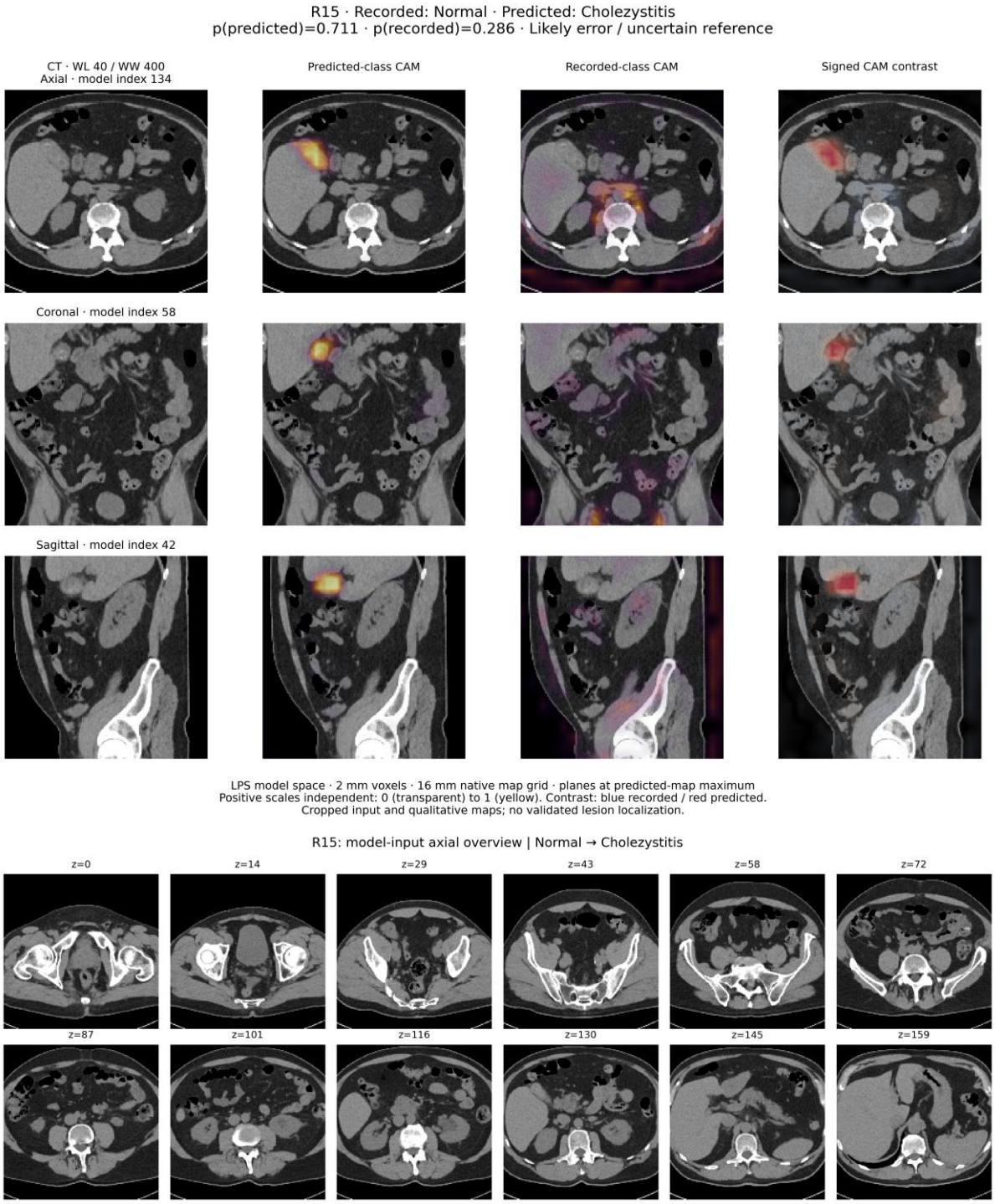


Occlusion Δp: targeted 0.053; random mean 0.012; minimum-intensity control 0.028. Fold-map correlation: 0.753; randomized-head correlation: 0.141.

R15 Likely error and uncertain reference

Original reference: Normal. FlexiCT prediction: Cholecystitis (probability 0.711; 4/5 fold votes).

Reviewer comment: Gallstones, chronic wall changes and symptoms improving with antibiotics; no definitive reference diagnosis.

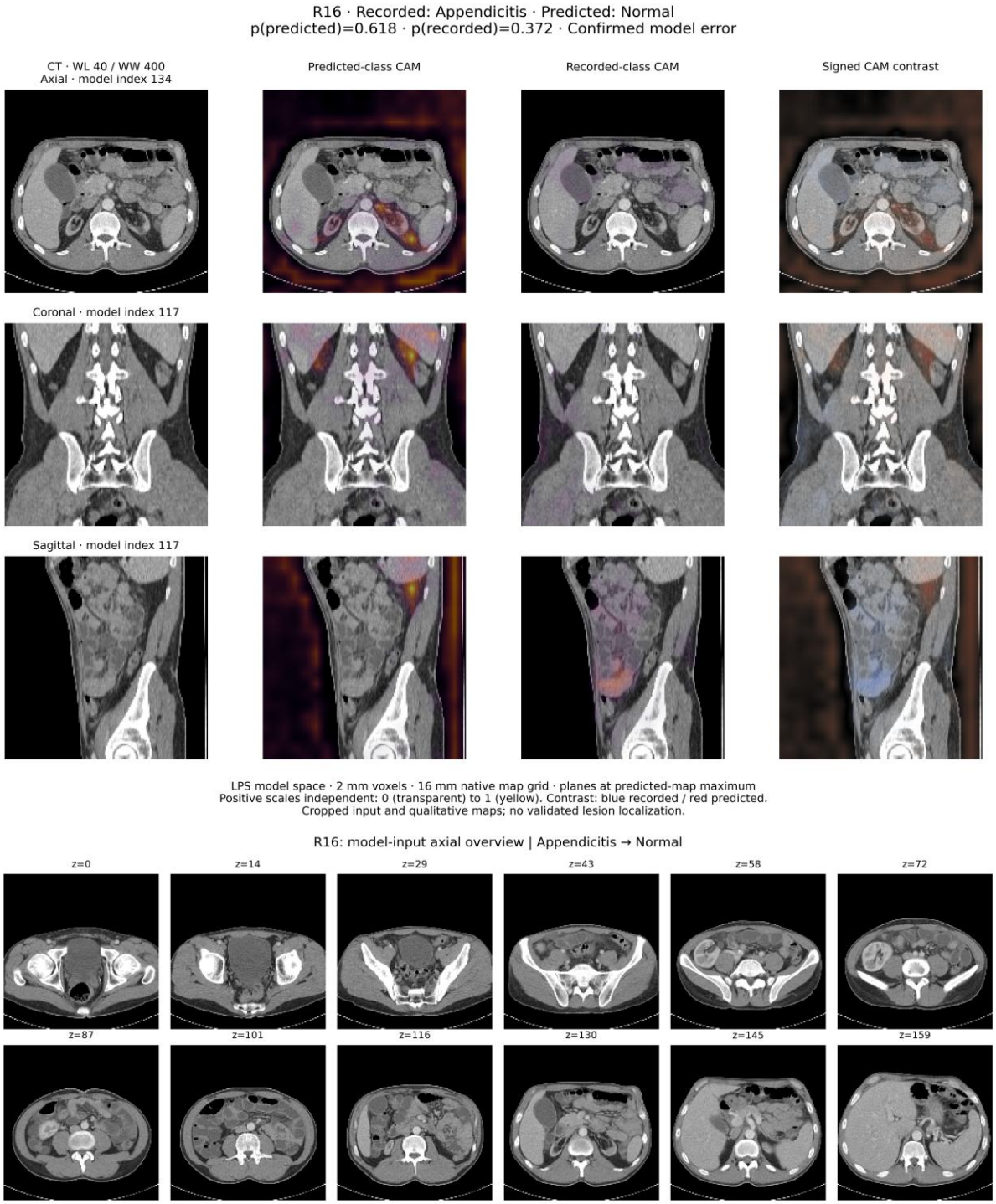


Occlusion Δp: targeted 0.709; random mean 0.386; minimum-intensity control 0.706. Fold-map correlation: 0.650; randomized-head correlation: 0.160.

R16 Confirmed model error

Original reference: Appendicitis. FlexiCT prediction: Normal (probability 0.618; 4/5 fold votes).

Reviewer comment: Uncomplicated appendicitis; anatomy altered by renal transplantation.

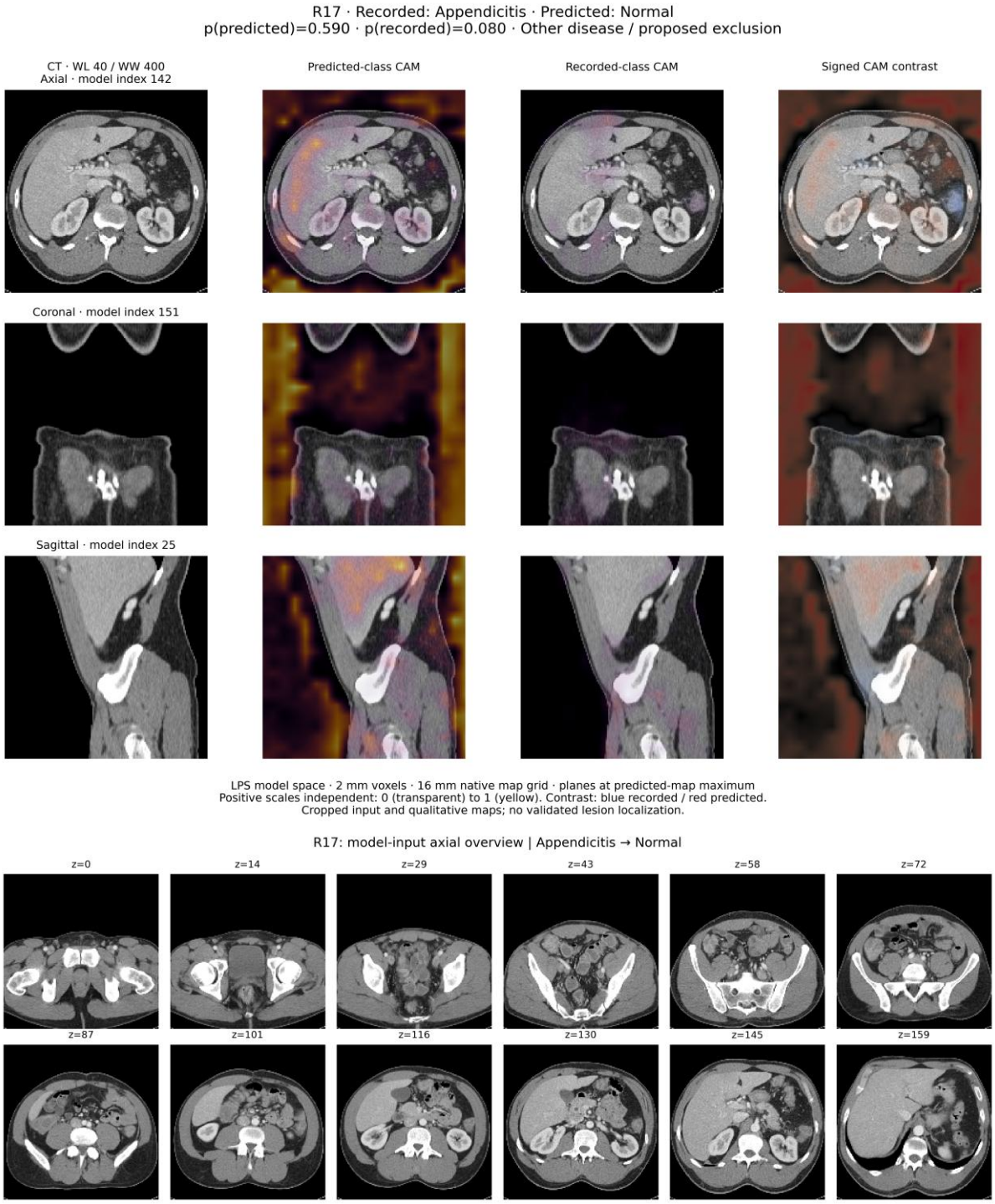


Occlusion Δp: targeted 0.024; random mean -0.189; minimum-intensity control -0.030. Fold-map correlation: 0.871; randomized-head correlation: -0.531.

R17 Other disease and proposed exclusion

Original reference: Appendicitis. FlexiCT prediction: Normal (probability 0.590; 3/5 fold votes).

Reviewer comment: Extensive ulcerative colitis with secondary appendiceal enhancement rather than primary appendicitis.

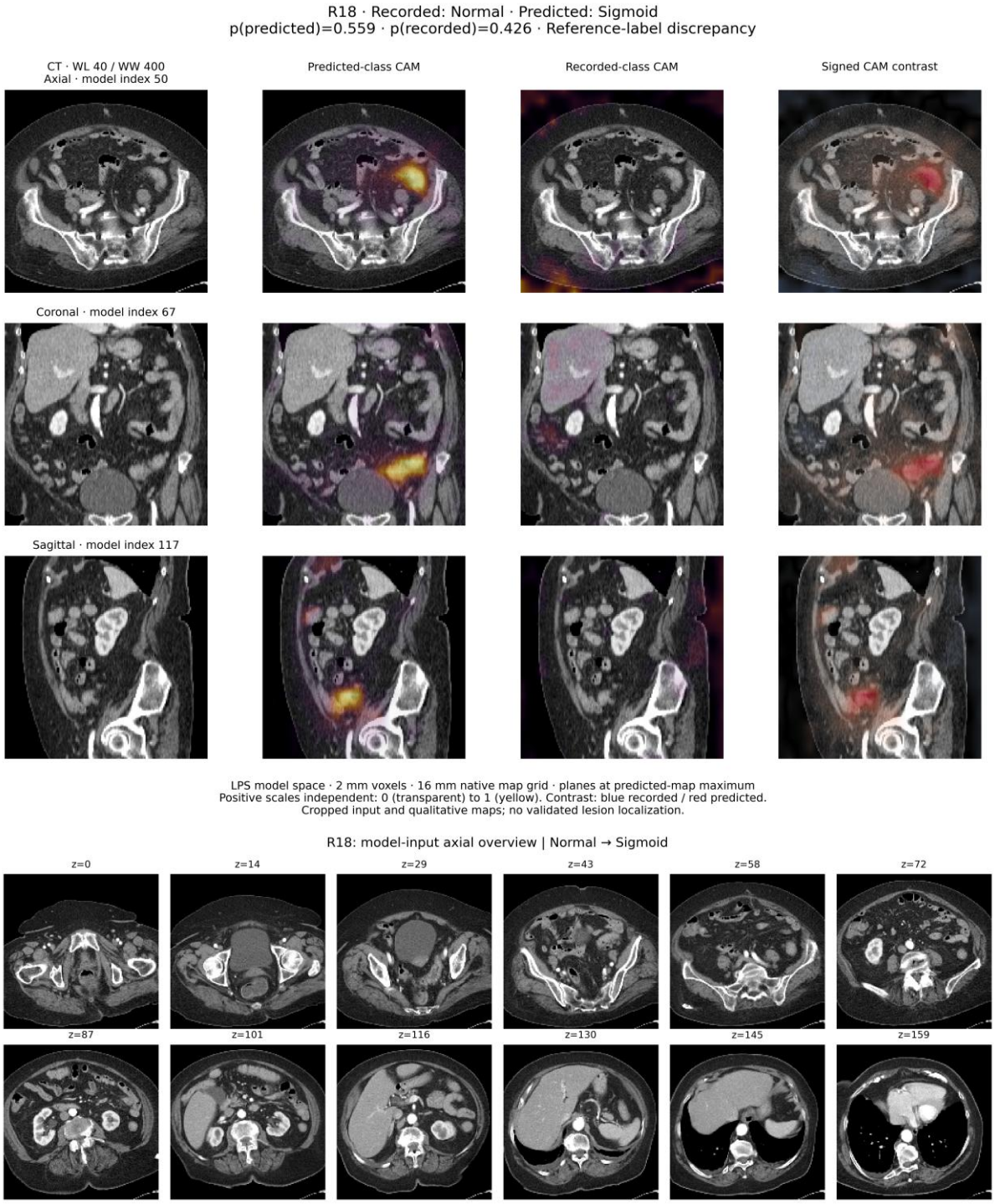


Occlusion Δp: targeted -0.085; random mean -0.124; minimum-intensity control 0.016. Fold-map correlation: 0.681; randomized-head correlation: -0.420.

R18 Reference-label discrepancy

Original reference: Normal. FlexiCT prediction: Sigmoid diverticulitis (probability 0.559; 3/5 fold votes).

Reviewer comment: Bacterial colitis with focal sigmoid inflammation and diverticula; diverticulitis considered likely.

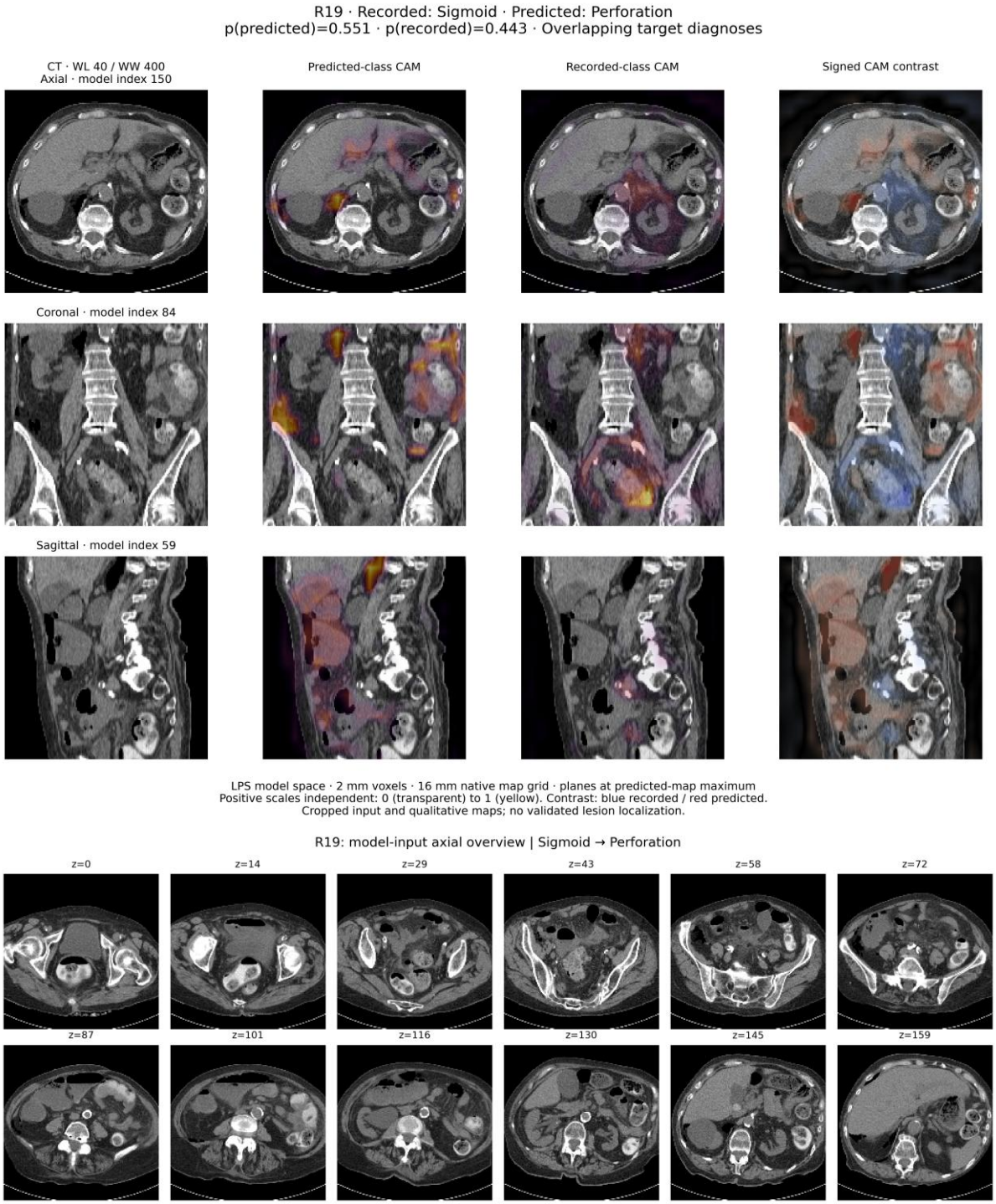


Occlusion Δp: targeted 0.555; random mean 0.376; minimum-intensity control 0.545. Fold-map correlation: 0.756; randomized-head correlation: 0.229.

R19 Overlapping target diagnoses

Original reference: Sigmoid diverticulitis. FlexiCT prediction: Perforation (probability 0.551; 3/5 fold votes).

Reviewer comment: Sigmoid diverticulitis with perforation.

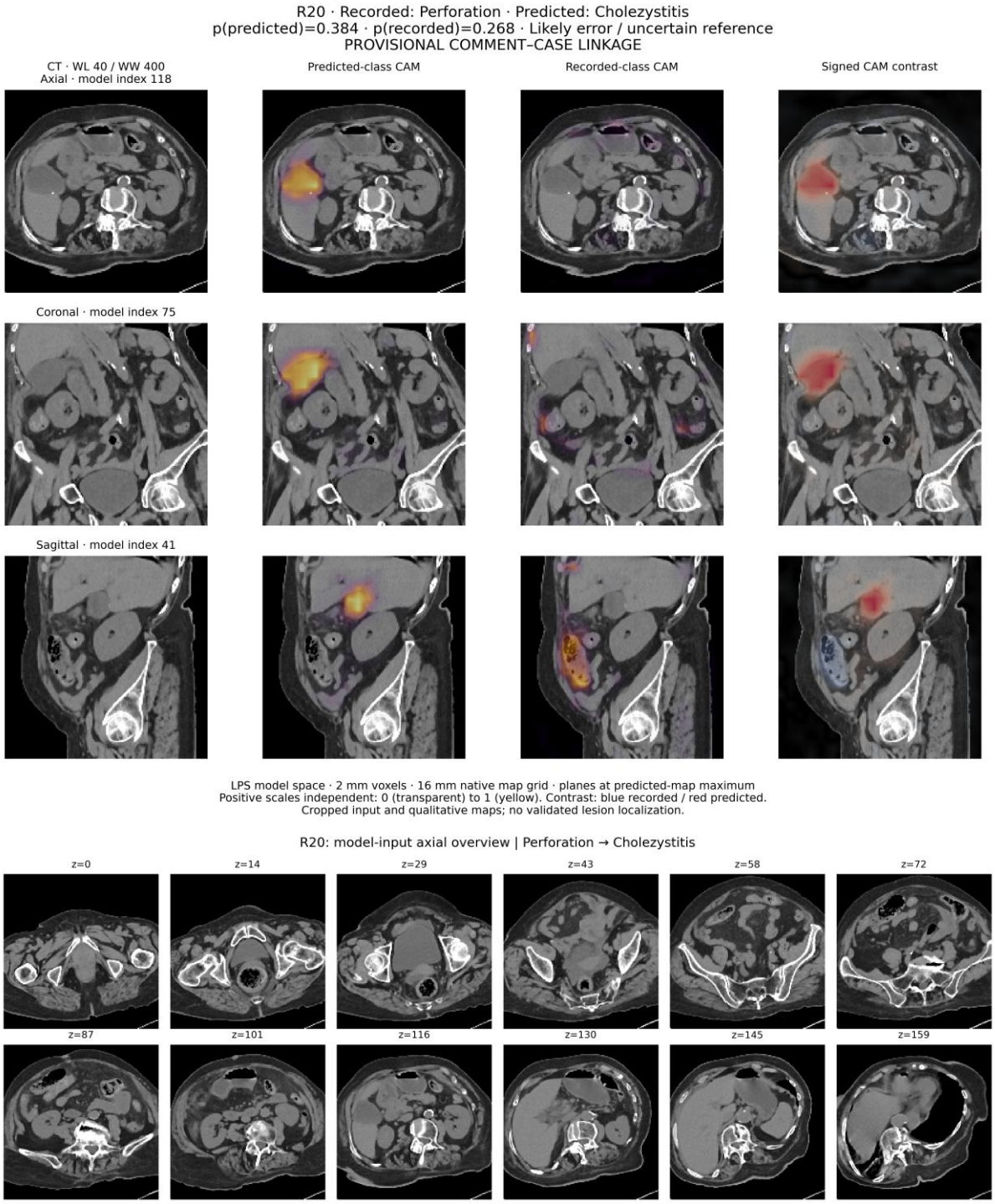


Occlusion Δp: targeted 0.152; random mean -0.049; minimum-intensity control 0.172. Fold-map correlation: 0.693; randomized-head correlation: -0.328.

R20 Likely error and uncertain reference

Original reference: Perforation. FlexiCT prediction: Cholecystitis (probability 0.384; 2/5 fold votes).

Provisional reviewer comment: Reviewer describes pyloric perforation with minimal free air and gallbladder hydrops mimicking cholecystitis; exclusion or discussion suggested.



Occlusion Δp: targeted 0.377; random mean 0.268; minimum-intensity control 0.379. Fold-map correlation: 0.460; randomized-head correlation: 0.169.